(a) Candidate Pool Budget K & Latency Trade-off

Candidate Recall@K (%)

Candidate Retrieval Latency (ms)

Candidate Pool Budget (K)

Pareto Knee ($K = 50$, 56.3%)

CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Latency (ms/sample)

(b) RRF Smoothing Parameter μ across City Densities

RRF Fused Top-10 Recall (%)

RRF Smoothing Constant (μ)

NYC Opt: $\mu = 5$ (61.0%)

CA Opt: $\mu = 50$ (42.0%)

CA (Sparse, Large-scale)
NYC (Compact, Dense)
TKY (High-density Network)

(c) Semantic Cluster Number K_c vs. Spatial Radius & Coverage

Mean Spatial Cluster Radius (km)

Non-singleton Bucket Coverage (%)

Semantic Cluster Vocabulary Capacity (K_c)

Optimal $K_c = 256$ (Radius ≈ 6.1km)

CA Mean Radius (km)
NYC Mean Radius (km)
TKY Mean Radius (km)
Bucket Coverage (%)

(d) Profile Token Length L_{prof} vs. Output Robustness

Check-in History Retention (%)

Valid JSON Parse Rate (%)

Max Profile Token Length (L_{prof})

Optimal $L_{prof} = 120$
(100% Parse, 48.6% History)

CA History Retention
NYC History Retention
TKY History Retention
Valid JSON Parse Rate (%)